Hybrid Deep Reinforcement Learning and PID Control for Robust Magnetic Catheter Tip Tracking

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***Abstract*—Minimally invasive surgery (MIS) facilitates complex interventions through small incisions, reducing patient's trauma and recovery time, with magnetically actuated catheters playing a pivotal role in precise navigation of confined anatomical pathways. These devices, embedding permanent magnets at the tip, enable remote steering via external fields, enhancing dexterity. A novel hybrid controller is presented, integrating deep deterministic policy gradient (DDPG) for wrench mapping with proportional-integral-derivative (PID) feedback to form a closed-loop system. Evaluation occurs under complex conditions, including varying trajectory tracking velocities and cumulative offset disturbances on different paths. Results indicate superiority of the hybrid model over standalone PID, with RMSE at the highest tracking speed of 2.21 mm versus 6.76 mm for PID, and on square trajectories under disturbances, reduced RMSE alongside lower overshoot of 1.50% and settling time of 0.64 s compared to PID's 4.91% and 0.72 s, thereby maintaining trajectory tracking across speed variations and demonstrating enhanced robustness.**

***Index Terms*—Magnetic catheter, Minimally invasive surgery, Deep reinforcement learning, Cosserat rod model, Soft robotics, Hybrid control**

I. Introduction

Minimally invasive surgery (MIS) is preferred over open methods, enabling complex interventions through small incisions and reducing patient's trauma and recovery time [1]. MIS instruments are typically miniaturized, flexible, and suited for confined anatomical navigation, with catheters and capsule-like devices as common examples. Endoscopic capsules enable navigation through complex paths while adapting to body contours [2] [3]. Catheters provide flexibility to protect tissues via even force distribution and access distant sites in confined routes [4] [5]. Given catheters' crucial role in MIS, this paper focuses on them. Magnetically guided instruments enhance MIS techniques. Over two decades, magneto-surgery using external magnetic fields has evolved into a versatile platform for catheter interventions [6]. A magnetic catheter embeds permanent magnets (often NdFeB) near its tip, actuated by external fields from magnets or coils [7].

The magnetic catheter system generates a moment aligning with an external magnetic field, enabling controlled tip bending and steering [8]. This setup applies force and torque inputs to the catheter tip. As Limpabandhu et al. reported, magnetically steerable catheters provide enhanced control and dexterity in MIS [9]. This remote precision addresses limitations of conventional catheters requiring manual twisting. Traditional methods rely on fluoroscopy for real-time X-ray visualization of catheter position, often laborious, trial-and-error-based, accuracy-limited, and radiation-exposing [10] [11]. Automated magnetic control alleviates these issues. Pratt et al. noted that autonomous continuum control achieves precision surpassing skilled surgeons [4]. External magnetic localization enables tracking without continuous radiation [12]. Overall, magnetic guidance improves safety by shortening procedure time, reducing radiation dose, and boosting clinical efficacy, as shown by platforms like Stereotaxis Niobe/Genesis [13].

Various classical and learning-based control approaches have been applied to magnetic catheters. Classical methods solve equilibrium equations or approximate segments [8], combined with feedback like PID or model-predictive control for path steering. For example, Sikorski et al. created a vision-based 3D system inverting the Cosserat model in real time to compute magnetic fields for desired tip poses [14]. Pratt et al. employed a model predictive controller and observers for sub-millimeter accuracy [4]. Others use finite-element or reduced-order models with PID or computed torque controllers, validated experimentally for trajectory following [15].

However, model-based controllers have limitations. Solving Cosserat equations is computationally intensive, needing iterative integration or finite-element methods per update [11]. Constraints, contacts, and uncertainties reduce performance, causing slow operation or simplifications that hinder accuracy and responsiveness.

Learning-based methods, particularly deep reinforcement learning (DRL), promise effective robotic control. DRL learns policies via trial and error in simulation, bypassing runtime model solving [16]. By defining objectives as reward functions, RL agents optimize performance through environmental interaction, implicitly capturing inverse behaviors without solving governing equations. Trained DRL agents directly output control actions like magnetic field vectors from desired targets, avoiding Cosserat dynamics resolution per tip pose. This approach succeeds in related fields; Yao et al. used DRL for magnetized robots in complex trajectories, solving inverse problems without guidance [16]. In continuum and soft robotics, RL yields closed-loop policies for compliant manipulators, skipping dynamic inversion and enhancing robustness to disturbances [17]. DRL also controls magnetic catheters with

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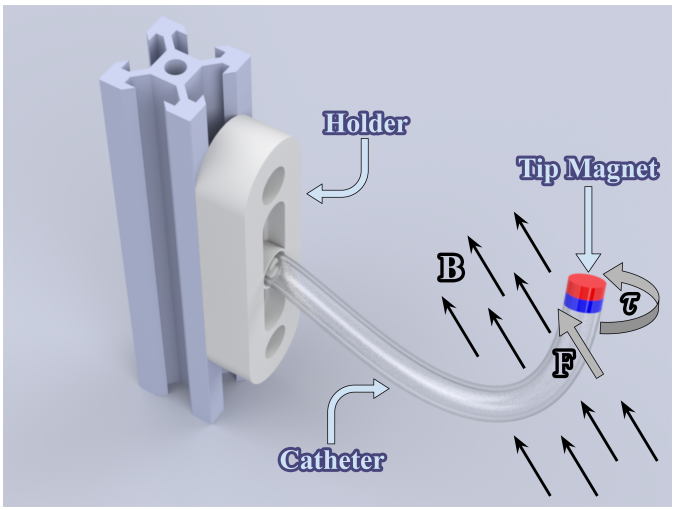


Fig. 1. Conceptual setup of the magnetic catheter, where external fields and gradients exert wrenches at the tip magnet for movement.

adaptive, model-bypassing strategies [18]. DRL's generalization and model-independence motivate its control application. However, open-loop DRL showed limited robustness to disturbances and perturbations.

To address these limitations, a hybrid DRL-PID strategy is introduced. The DRL component trains offline to produce nonlinear feedforward magnetic commands for complex tip motions, effectively inverting the Cosserat model, while the PID adds real-time closed-loop corrections to ensure robustness. This hybrid approach achieves accurate trajectory tracking and superior disturbance rejection.

II. Catheter Model & Parameters

A. Catheter Properties and Assumptions

To derive the governing equations and design the controller, the catheter is modeled with one end fixed and the other end free to move in space, actuated by forces and torques applied at its tip through magnetic actuation. The conceptual model of the defined catheter and its setup is illustrated in Figure 1.

As we have access to a fabricated PDMS catheter for this purpose, our simulations and conceptual catheter are based on the same properties as the available catheter, which are listed in Table I.

TABLE I
Catheter Parameters

Symbol	Description	Value	Unit
L	Catheter length	0.124025	m
A	Cross-sectional area	1.6908×10^{-5}	m^2
R	Outer radius	0.003385	m
r	Inner radius	0.002465	m
ρ	Material density	1030	kg/m^3
E	Young's modulus	1.888567×10^6	Pa
G	Shear modulus	6.638528×10^5	Pa
ν	Poisson's ratio	0.422429	-
$\mathbf{g}$	Gravity	9.81	m/s^2
$\mathbf{Mg}$	Tip magnet mass	0.011	kg

The stiffness matrix for shear and extension is defined as

$$\mathbf{K_{se}} = \begin{bmatrix} GA & 0 & 0 \\ 0 & GA & 0 \\ 0 & 0 & EA \end{bmatrix}.$$

The stiffness matrix for bending and twisting is defined as

$$\mathbf{K_{bt}} = \begin{bmatrix} EI_{xx} & 0 & 0 \\ 0 & EI_{yy} & 0 \\ 0 & 0 & GI_{zz} \end{bmatrix}.$$

The I terms represent the second moment of inertia of the cylindrical catheter. The damping matrices $\mathbf{B_{se}}$ and $\mathbf{B_{bt}}$ are 10% of $\mathbf{K_{se}}$ and $\mathbf{K_{bt}}$ matrices.

B. Dynamic Model

A rigorous mathematical model is essential for describing the kinematics and dynamics of the magnetic catheter. The dynamic Cosserat rod model is employed to capture large deformations and spatially varying curvature, providing a geometrically exact representation of the catheter's shape under magnetic actuation. The governing equations follow as the foundation for controller design.

The governing PDEs of the Cosserat rod model involve derivatives with respect to time (t) and arc length (s). The foundational equations governing the evolution of position and orientation are:

$$\frac{\partial \mathbf{P}}{\partial s} = \mathbf{R}\mathbf{v} \qquad \frac{\partial \mathbf{P}}{\partial t} = \mathbf{R}\mathbf{q} \tag{1}$$

$$\frac{\partial \mathbf{R}}{\partial s} = \mathbf{R} \times \mathbf{u} \qquad \frac{\partial \mathbf{R}}{\partial t} = \mathbf{R} \times \omega \tag{2}$$

where $\mathbf{P}$ and $\mathbf{R}$ are the section position vector and rotation matrix in global frame, and $\mathbf{v}$ and $\mathbf{u}$ are the rate of change of position with respect to arclength and curvature vector in the local frame.

These are complemented by the balance laws for linear and angular momentum:

$$\frac{\partial \mathbf{n}}{\partial s} = \rho A \mathbf{R}\left(\omega \times \mathbf{q} + \frac{\partial \mathbf{q}}{\partial t}\right) - \mathbf{f} \tag{3}$$

$$\frac{\partial \mathbf{m}}{\partial s} = \frac{\partial}{\partial t}\left(\mathbf{R}\rho\mathbf{J}\omega\right) - \frac{\partial \mathbf{P}}{\partial s} \times \mathbf{n} - \mathbf{l} \tag{4}$$

where $\mathbf{n}$ and $\mathbf{m}$ denote internal force and moment in the global frame, $\mathbf{q}$ and ω local-frame velocity and angular velocity, and $\mathbf{f}$ and $\mathbf{l}$ distributed force and torque along the catheter.

Additionally, compatibility conditions ensure consistency between velocities and strains:

$$\frac{\partial \mathbf{q}}{\partial s} = \frac{\partial \mathbf{v}}{\partial t} - \mathbf{u} \times \mathbf{q} + \omega \times \mathbf{v} \tag{5}$$

$$\frac{\partial \omega}{\partial s} = \frac{\partial \mathbf{u}}{\partial t} - \mathbf{u} \times \omega \tag{6}$$

The system features independent variables $\mathbf{v}$ and $\mathbf{u}$, which are linked to internal forces and moments via a linear elastic constitutive law incorporating material damping:

$$\mathbf{n} = \mathbf{R}\left[\mathbf{K_{se}}\left(\mathbf{v}-\mathbf{v}^{*}\right)+\mathbf{B_{se}}\frac{\partial \mathbf{v}}{\partial t}\right] \tag{7}$$

$$\mathbf{m} = \mathbf{R}\left[\mathbf{K_{bt}}\left(\mathbf{u}-\mathbf{u}^{*}\right)+\mathbf{B_{bt}}\frac{\partial \mathbf{u}}{\partial t}\right] \tag{8}$$

where $\mathbf{v}^*$ is the undeformed strain vector and $\mathbf{u}^*$ is the undeformed curvature vector.

This setup forms a boundary value problem (BVP), where boundary conditions—such as fixed proximal ends and actuated distal tips—must be specified to solve for the catheter's spatiotemporal behavior, enabling predictive simulations of its response to magnetic fields.

For numerical tractability, the partial differential equations are discretized using a generalized implicit time-stepping scheme with parameter α to modulate linearization:

$$\mathbf{y}_t^{(i)} = c_0\mathbf{y}^{(i)} + c_1\mathbf{y}^{(i-1)} + c_2\mathbf{y}^{(i-2)} + d_1\mathbf{y}_t^{(i-1)} \tag{9}$$

Where the coefficients are defined as:

$$c_0 = \frac{(1.5+\alpha)}{[\delta t(1+\alpha)]} \qquad c_1 = -\frac{2}{\delta t}$$
$$c_2 = \frac{(0.5+\alpha)}{[\delta t(1+\alpha)]} \qquad d_1 = \frac{\alpha}{(1+\alpha)}$$

Second-order accuracy holds for $\alpha \in [-0.5, 0]$. Within this spectrum, the trapezoidal rule ($\alpha = -0.5$) offers a harmonious blend of stability and precision, whereas the second-order backward differentiation formula ($\alpha = 0$) prioritizes robustness for stiff nonlinear systems. Accordingly, $\alpha = 0$ is adopted here.

Post-discretization, the time derivatives simplify to:

$$\frac{\partial \mathbf{v}}{\partial t} = c_0\mathbf{v} + \mathbf{v^h} \qquad \frac{\partial \mathbf{u}}{\partial t} = c_0\mathbf{u} + \mathbf{u^h} \tag{10}$$

$$\frac{\partial \mathbf{q}}{\partial t} = c_0\mathbf{q} + \mathbf{q^h} \qquad \frac{\partial \mathbf{w}}{\partial t} = c_0\omega + \omega^{\mathbf{h}} \tag{11}$$

By substituting these values into equations (1) to (8), we reach the solvable catheter equations.

C. Quasi-Static Model

To train the DRL agent, a mapper connects it to the dynamic model using Cosserat static equations, derived by eliminating time-dependent terms from the dynamic counterparts for stability and physical grounding. This ensures robust policy learning.

The core kinematic relations simplify to:

$$\frac{\partial \mathbf{P}}{\partial s} = \mathbf{R}\mathbf{v} \tag{12}$$
$$\frac{\partial \mathbf{R}}{\partial s} = \mathbf{R}\times\mathbf{u}$$

Momentum balances under static conditions become:

$$\frac{\partial \mathbf{n}}{\partial s} = -\rho A\mathbf{g} - \mathbf{f} \tag{13}$$

$$\frac{\partial \mathbf{m}}{\partial s} = -\frac{\partial \mathbf{P}}{\partial s}\times\mathbf{n} - \mathbf{l} \tag{14}$$

Constitutive laws link strains to internal loads without damping or inertial effects:

$$\mathbf{v} = \mathbf{v}_0 + \mathbf{K_{se}^{-1}}\mathbf{R}^T\mathbf{n} \tag{15}$$

$$\mathbf{u} = \mathbf{u}_0 + \mathbf{K_{bt}^{-1}}\mathbf{R}^T\mathbf{m} \tag{16}$$

Specifying end-effector wrenches at the distal tip, with proximal constraints, yields the steady-state shape, illuminating how magnetic fields sculpt the catheter's posture for path planning.

III. Controller Design

A. DRL Controller

Given the nature of the catheter model, a deterministic policy is required for precise, repeatable actuation. Thus, the Deep Deterministic Policy Gradient (DDPG) algorithm is employed, an actor-critic method ideal for continuous action spaces. DDPG uses an actor network to produce deterministic actions from states and a critic to estimate expected rewards, enabling joint optimization via off-policy learning and experience replay.

The DDPG is configured to map desired tip positions to wrenches in steady-state, with state space as $\mathbf{p}_d \in \mathbb{R}^{3\times1}$ and action space as $\mathbf{w} \in \mathbb{R}^{6\times1}$ (forces and torques). During training, the actor's wrench is applied to the quasi-static model to obtain $\mathbf{p}_a$, training a steady-state mapper for control. Figure 2 depicts the architecture, where desired position inputs the actor, wrench feeds the model for achieved position, and actor-critic interact in policy optimization.

Training spans 500,000 steps with random reachable tip positions for generality and robustness.

The neural network architectures and hyperparameters are in Table II; both use fully connected layers with ReLU. DDPG parameters are in Table III.

The reward function, derived from RMSE between desired and achieved positions (RMSE $= \sqrt{\frac{1}{n}\sum_{i=1}^{n} e_i^2}$), is $r = \exp(-k\cdot\text{RMSE})$ with $k = 30$ for sensitivity and stability. This bounds rewards in [0,1], favoring small errors for convergence and precise positioning.

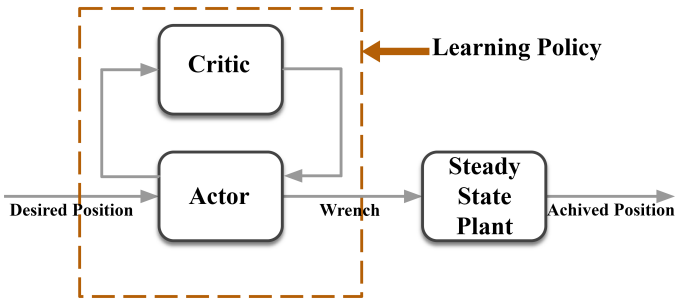


Fig. 2. Open-loop DDPG training logic for the steady-state mapper.

TABLE II
NETWORK ARCHITECTURE AND LEARNING HYPERPARAMETERS

Parameter	Value
Actor Hidden Layers	(400, 300)
Critic Hidden Layers	(400, 300)
Actor Learning Rate	1×10^{-4}
Critic Learning Rate	2×10^{-4}
Batch Size	256

TABLE III
DDPG HYPERPARAMETERS

Parameter	Value
Discount Factor (γ)	0.99
Soft Update Rate (τ)	0.005
Noise Decay	0.99
Initial Noise Scale	0.5
Minimum Noise Scale	0.01
Training Steps	500,000

B. PID Controller

As a baseline for comparison, a proportional-integral-derivative (PID) controller is employed, representing a conventional and widely used approach in engineering applications for feedback control. The PID adjusts the control inputs based on the error between the desired and actual tip positions in a closed-loop configuration.

The position error is defined as the 3×1 vector $\mathbf{e} = \mathbf{p}_d - \mathbf{p}_a$, where $\mathbf{p}_d$ and $\mathbf{p}_a$ denote the desired and achieved tip positions, respectively. To generate the 6×1 wrench vector, two separate PID controllers are utilized: one for the 3×1 force components and another for the 3×1 torque components, both driven by the position error. The PID control law is given by:

$$\mathbf{F} = K_{p1}\mathbf{e} + K_{i1}\int \mathbf{e}\,dt + K_{d1}\frac{d\mathbf{e}}{dt},$$
$$\mathbf{T} = K_{p2}\mathbf{e} + K_{i2}\int \mathbf{e}\,dt + K_{d2}\frac{d\mathbf{e}}{dt}, \quad (17)$$

where $\mathbf{F}$ and $\mathbf{T}$ are the force and torque outputs; K_{p1}, K_{i1}, K_{d1} are gains for forces; K_{p2}, K_{i2}, K_{d2} for torques.

The gains are empirically tuned on the dynamic catheter model to minimize tracking error and ensure stability under nominal trajectories and disturbances.

C. Hybrid DRL-PID Controller

To harness the strengths of both classical and learning-based controllers, we propose a hybrid framework that integrates the DRL mapper with PID feedback, creating a robust closed-loop system for catheter tip control. While the DRL excels at predicting wrenches for precise trajectory following, it operates open-loop and falters under disturbances. In contrast, PID provides real-time error correction but may lack sophistication for intricate paths. By blending them, we achieve a synergistic controller that combines DRL's predictive accuracy with PID's disturbance rejection, enhancing overall reliability in dynamic environments.

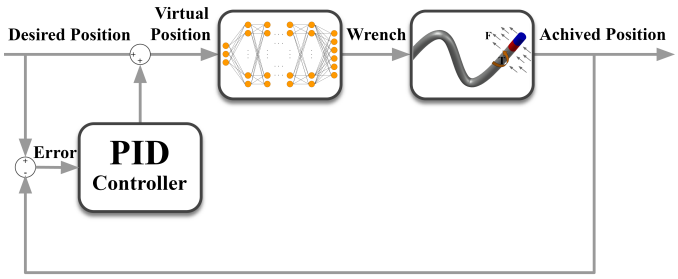


Fig. 3. Schematics of the closed-loop hybrid DRL-PID control model.

The trained actor approximates the inverse quasi-static mapping from desired tip position to equilibrium wrench. During dynamic tracking, PID feedback modifies the actor's position input to compensate for tracking errors. Specifically, the PID processes the position error $\mathbf{e} = \mathbf{p}_d - \mathbf{p}_a$ to generate a corrective offset $\Delta\mathbf{p}$, yielding the virtual target $\mathbf{p}_v = \mathbf{p}_d + \Delta\mathbf{p}$. The actor maps this virtual target to the wrench applied to the dynamic catheter model. Figure 3 illustrates the resulting closed-loop architecture.

The PID gains in this hybrid setup are empirically tuned on the dynamic catheter equations to minimize tracking errors and maintain stability across nominal trajectories and disturbances.

IV. SIMULATION RESULTS

A. DRL Training Performance

The efficacy of the DRL agent, trained via DDPG over 500,000 steps, is shown through learning curves for actor/critic losses, episode returns, and success rate, as in Figure 4.

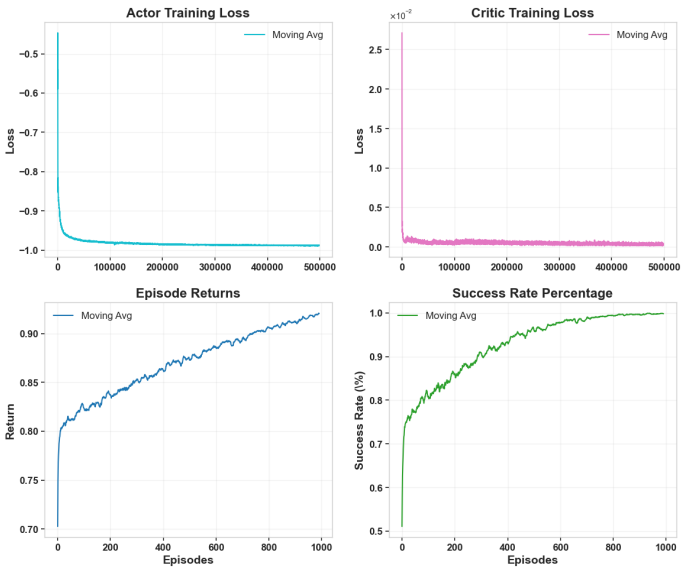


Fig. 4. Learning curves for the DDPG-trained DRL agent: (top-left) actor loss, (top-right) critic loss, (bottom-left) episode returns, (bottom-right) success rate per episode (500 steps per episode).

The actor and critic losses experience a rapid decrease, enabling the model to identify the positioning policy by step 10,000 and stabilize near -0.95 and 0, with moving averages

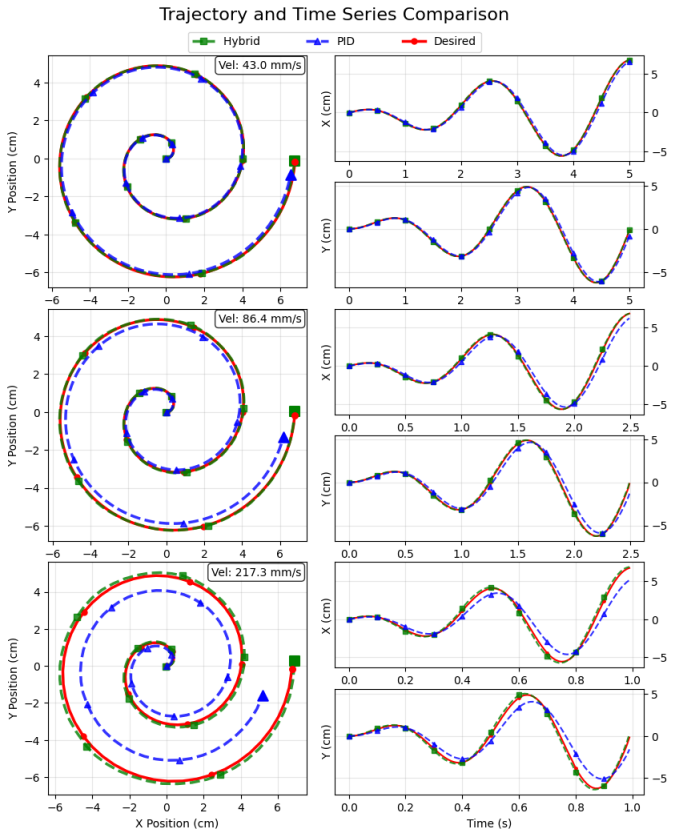


Fig. 5. Left: Top-down views of trajectory tracking on the spiral path at 43 mm/s, 86 mm/s, and 217 mm/s; Right: x-y trajectory tracking versus time.

over 100 steps confirming convergence and indicating an effective learned policy. Episode returns and success rates increase during training, with the mean reward approaching 0.95 and the success rate approaching 100%. Success is defined as the fraction of positioning steps per episode for which the instantaneous RMSE across the three position coordinates is below $10\,\mathrm{mm}$. The $10\,\mathrm{mm}$ threshold was used to summarize training success and does not represent a clinically validated positioning tolerance.

B. Trajectory Tracking Performance

To compare the trajectory tracking performance of the hybrid DRL-PID and standalone PID controllers, both were tuned on a spiral trajectory at nominal speed (43 mm/s). Evaluation was then conducted on the same trajectory at accelerated speeds of 86 mm/s and 217 mm/s, to simulate demanding vascular navigation scenarios. The top-down views of the achieved tip paths and temporal dynamics are shown in Figure 5.

As observed in the left panels, with increasing tracking speed, the PID controller develops noticeable deviations behind the desired path. In contrast, the hybrid controller maintains closer alignment to the reference trajectory with lower error than PID standalone, leveraging the DRL's predictive mapping augmented by PID corrections to minimize lag.

TABLE IV
TOTAL RMSE AT VARYING TRACKING SPEEDS (MM)

Controller	43 mm/s	86 mm/s	217 mm/s
Hybrid	0.492	0.968	2.207
PID	2.010	3.331	6.755

Quantitative comparison of the total root mean square error across all axes is provided in Table IV. The hybrid consistently achieves lower errors, with RMSE increasing modestly from 0.49 mm at 43 mm/s to 2.21 mm at 217 mm/s, compared to the PID's sharper rise from 2.01 mm to 6.76 mm.

The right panels of Figure 5 provide a detailed view of the temporal dynamics, plotting x-y trajectories versus time. At higher speeds, the PID response exhibits a slight phase change relative to the desired path, attributable to its tuning on the nominal trajectory and requiring retuning for faster dynamics. Conversely, the hybrid controller delivers more accurate outputs adapted to changing conditions via the trained mapper, enabling the tip to remain proximate to the desired points throughout.

C. Disturbance Rejection

To evaluate the hybrid controller's robustness under disturbances, its performance was assessed alongside the standalone PID on square and circle trajectories with cumulative offset disturbances. As illustrated in Figure 6, four constant $5\,\mathrm{mN}$ forces normal to the path in the x-y plane were added to the controller outputs when the tip came within $10\,\mathrm{mm}$ of successive markers. Each persisted until the run ended, accumulating vectorially. This setup enables a direct comparison of the controllers' disturbance rejection capabilities.

Performance metrics for both trajectories are summarized in Table V.

These results demonstrate that the hybrid controller exhibits greater robustness to disturbances compared to the standalone PID across both trajectories. It achieves lower overshoot, indicating enhanced resilience to perturbations, and a marginal improvement in settling time of approximately 0.08 s on average. Furthermore, the hybrid's RMSE is notably reduced, underscoring its superior precision under disturbances. This advantage arises from the mapper, which generates wrenches informed by the desired position and implicit dynamics awareness, producing more robust outputs than the PID, which lacks such learned contextual adaptation.

TABLE V
PERFORMANCE METRICS UNDER CUMULATIVE OFFSET DISTURBANCES

Controller	Hybrid		PID	
Trajectory	Square	Circle	Square	Circle
RMSE (mm)	0.627	0.574	1.384	1.320
Settling Time (s)	0.64	0.54	0.72	0.61
Overshoot (%)	1.50	1.72	4.91	5.81

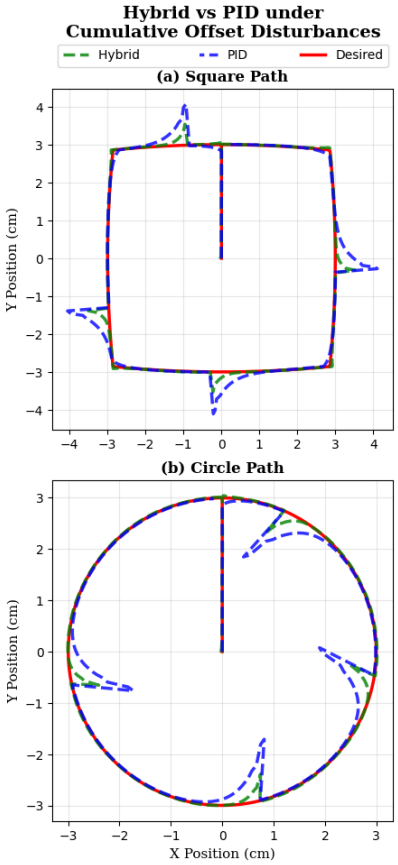


Fig. 6. Disturbance rejection under cumulative offset forces on (a) square and (b) circle trajectories: Desired (dashed), PID (blue), hybrid DRL-PID (green).

V. Conclusion

This study introduces a hybrid DRL-PID controller for magnetically actuated catheters, synergizing deep reinforcement learning and classical control to enhance trajectory tracking and robustness. By employing DDPG as a steady-state mapper, the DRL component delivers precise wrench predictions for complex paths, complemented by PID feedback for real-time error correction and disturbance rejection. Simulation results demonstrate the hybrid's superiority over standalone PID in trajectory tracking on the tuned spiral path at accelerated speeds, with RMSE rising modestly from 0.49 mm to 2.21 mm across tracking speeds of 43 to 217 mm/s, compared to PID's sharp escalation from 2.01 mm to 6.76 mm. Under cumulative offset disturbances, the hybrid achieves lower RMSE of 0.63 mm on the square path and 0.57 mm on the circle path, alongside reduced overshoot of 1.50% on the square and 1.72% on the circle, as well as shorter settling times of 0.64 s on the square and 0.54 s on the circle—versus PID's RMSE of 1.38 mm and 1.32 mm, overshoot of 4.91% and 5.81%, and settling times of 0.72 s and 0.61 s, confirming enhanced robustness. Future work includes actuator implementation, experimental validation on prototypes and integration of advanced DRL variants like SAC or TD3 for fully autonomous navigation.

References

[1] D. V. Kladko and V. V. Vinogradov, "Magnetosurgery: principles, design, and applications," *Smart Materials in Medicine*, vol. 5, no. 1, pp. 24–35, 2024.

[2] M. Jamali, M. Z. Khameneh, M. Sarkhosh, and H. N. Pishkenari, "Enhanced Optimal Control of a Magnetically Actuated Capsule Endoscope Using Electromagnetic Coils Considering Small Intestine Dynamics," in *2024 12th RSI International Conference on Robotics and Mechatronics (ICRoM)*, pp. 125–130, 2024.

[3] M. R. Z. Khameneh, Z. Tolouei, M. Yousefi, H. N. Pishkenari, and A. Alasty, "Development and Experimental Evaluation of an Endoscopic Capsule Robot Equipped With a Fail-safe Biopsy Mechanism," in *2024 12th RSI International Conference on Robotics and Mechatronics (ICRoM)*, pp. 034–039, 2024.

[4] R. L. Pratt, B. E. Suesser, and A. J. Petruska, "Observed control of magnetic continuum devices," *Robotics*, vol. 12, no. 1, p. 11, 2023.

[5] J. Yang, Y. Zhang, Z. Liu, S. Huang, X. Huang, Y. Wang, M. Li, S. Zheng, F. Chen, J. Liu, et al., "Magnetically actuated multimodal bioelectronic catheter for minimally invasive surgery and sensing," *Nature Materials*, pp. 1–13, 2025.

[6] A. M. E. Noten, T. Szili-Torok, S. Ernst, D. Burkhardt, D. Cavaco, X. Chen, J. W. Cheung, C. de Chillou, E. Crystal, D. H. Cooper, et al., "Best practices in robotic magnetic navigation-guided catheter ablation of cardiac arrhythmias, a position paper of the Society for Cardiac Robotic Navigation," *Frontiers in cardiovascular medicine*, vol. 11, p. 1431396, 2024.

[7] H. Shen, S. Cai, Z. Wang, Z. Ge, and W. Yang, "Magnetically driven microrobots: Recent progress and future development," *Materials & Design*, vol. 227, p. 111735, 2023.

[8] C. Heunis, J. Sikorski, and S. Misra, "Flexible instruments for endovascular interventions: Improved magnetic steering, actuation, and image-guided surgical instruments," *IEEE robotics & automation magazine*, vol. 25, no. 3, pp. 71–82, 2018.

[9] C. Limpabandhu, Y. Hu, H. Ren, W. Song, and Z. T. H. Tse, "Magnetically steerable catheters: State of the art review," *Proceedings of the Institution of Mechanical Engineers, Part H: Journal of Engineering in Medicine*, vol. 237, no. 3, pp. 297–308, 2023.

[10] Y. Kim, E. Genevriere, P. Harker, J. Choe, M. Balicki, R. W. Regenhardt, J. E. Vranic, A. A. Dmytriw, A. B. Patel, and X. Zhao, "Telerobotic neurovascular interventions with magnetic manipulation," *Science Robotics*, vol. 7, no. 65, p. eabg9907, 2022.

[11] D. Tong, Z. Hao, J. Li, B. Sun, M. Liu, L. Wang, and W. Huang, "Real-time simulation enabled navigation control of magnetic soft continuum robots in confined lumens," *Journal of the Mechanics and Physics of Solids*, p. 106198, 2025.

[12] M. R. Z. Khameneh, M. Jamali, and H. N. Pishkenari, "Symmetry-based Independent Fully Convolutional Neural Networks for Magnetic Localization Using Sensor Array," *IEEE Sensors Journal*, 2025.

[13] J. Hwang, J. Kim, and H. Choi, "A review of magnetic actuation systems and magnetically actuated guidewire-and catheter-based microrobots for vascular interventions," *Intelligent Service Robotics*, vol. 13, no. 1, pp. 1–14, 2020.

[14] J. Sikorski, A. Denasi, G. Bucchi, S. Scheggi, and S. Misra, "Vision-based 3-D control of magnetically actuated catheter using BigMag—An array of mobile electromagnetic coils," *IEEE/ASME transactions on mechatronics*, vol. 24, no. 2, pp. 505–516, 2019.

[15] S. Djeffal, A. Ghoul, M. R. Morakchi, C. Mahfoudi, and M. Belkedari, "Optimized computer torque control and dynamic model of a spatial single section continuum robot," *Results in Control and Optimization*, vol. 12, p. 100264, 2023.

[16] J. Yao, Q. Cao, Y. Ju, Y. Sun, R. Liu, X. Han, and L. Li, "Adaptive actuation of magnetic soft robots using deep reinforcement learning," *Advanced Intelligent Systems*, vol. 5, no. 2, p. 2200339, 2023.

[17] T. G. Thuruthel, E. Falotico, F. Renda, and C. Laschi, "Model-based reinforcement learning for closed-loop dynamic control of soft robotic manipulators," *IEEE Transactions on Robotics*, vol. 35, no. 1, pp. 124–134, 2018.

[18] A. Centurelli, L. Arleo, A. Rizzo, S. Tolu, C. Laschi, and E. Falotico, "Closed-loop dynamic control of a soft manipulator using deep reinforcement learning," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 4741–4748, 2022.